Experiment 6: Breadth First Search (BFS)

Aim

To implement the Breadth First Search (BFS) algorithm in Python for graph traversal.

Objective

- To traverse all vertices level by level.
- To find the shortest path in an unweighted graph.

Flowchart

```
Start
     |
Create Graph
     |
Choose Start Node
     |
Add Node to Queue
     |
Queue Empty?
    / \
No      Yes
 |          |
Remove Node
 |
Visit Adjacent Nodes
 |
Add Unvisited Nodes
 |
Repeat
 |
End
```

Algorithm

1. Create a graph using an adjacency list.
2. Select the starting node.
3. Add the starting node to the queue.
4. Mark it as visited.
5. Remove a node from the queue.
6. Visit all unvisited adjacent nodes.
7. Add them to the queue.
8. Repeat until the queue becomes empty.

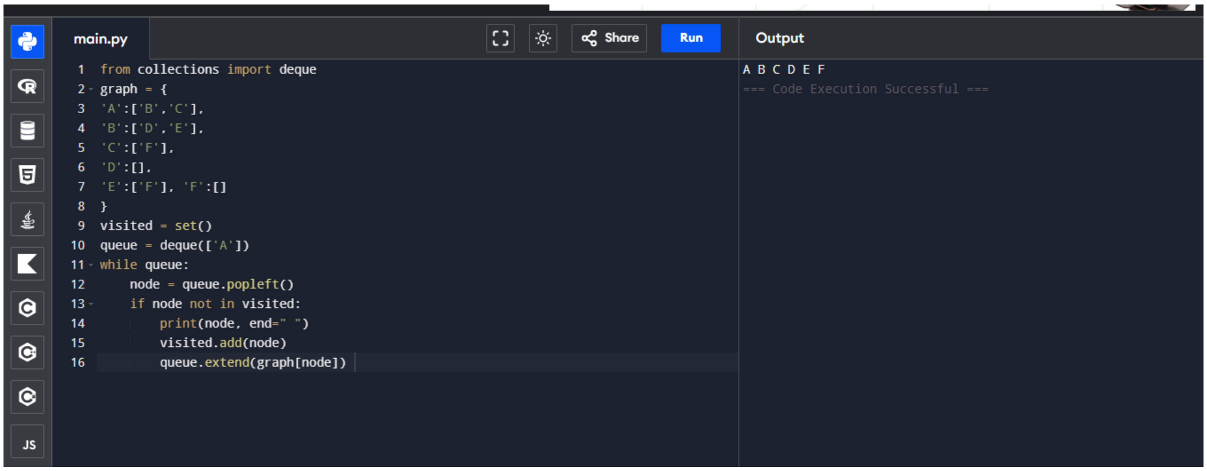
Program

```
from collections import deque graph = {
	'A':['B','C'],
	'B':['D','E'],
	'C':['F'],
	'D':[],
	'E':['F'], 'F':[]
}

visited = set() queue =
deque(['A'])

while queue:
	node = queue.popleft() if node
	not in visited:
		print(node, end=" ")
		visited.add(node)
		queue.extend(graph[node])
```

Output



Result

The BFS traversal was performed successfully.

Conclusion

The Breadth First Search algorithm visits nodes level by level using a queue.